Executive Summary: Parkinson's Disease Knowledge Graph Analysis

This report integrates multi-omics prioritization (Module 2) and machine learning druggability predictions (Module 3) with phenotype, pathway, drug, variant, and synthetic patient data into a Neo4j knowledge graph. By shifting from tabular relational joins to a property-graph schema, this approach enables multi-hop traversal to propose candidate disease mechanisms, rank candidate targets via network centrality, and evaluate cross-disease drug repurposing pathways.

Interpretation

1. Schema and Evidence Architecture

Property-Graph Schema Integration

I chose a property-graph schema because answering a multi-hop question with relational joins requires a separate linking table and JOIN per hop, which makes the query brittle, hard to read, and slow to rewrite when the question changes. A property graph allows nodes (entities) and edges (relationships) to store rich, context-specific metadata—such as association scores, variant p-values, and clinical trial phases—directly on the network structure. This design enables Cypher queries to express a five-hop question (Drug → Gene → Disease → Phenotype ← Disease) as a single readable pattern rather than a five-way JOIN.

Layer	Nodes (n	Edge type (n)	Evidence on edge	Source
Gene	16	ASSOCIATED_WITH	p-value, −log10 p, genome-wide vs suggestive, Open Targe	Modules 2-3; Open Targets
Variant	15	OCCURS_IN (15)	GWAS p-value, PubMed ID, positional mapping	GWAS Catalog (Module 2)
Pathway	15	PARTICIPATES_IN (2	Reactome ID	Reactome via Open Targets
Phenotyp	7	HAS_PHENOTYPE (1	HPO ID	HPO via Open Targets
Drug	8	TARGETS (8)	mechanism; primary target vs polypharmacology	ChEMBL via Open Targets
Patient	12	CARRIES (7)	risk-allele count	Synthetic cohort
Disease	3	–	MONDO ID	MONDO
Total	76	81		

Graph Metrics: Total Nodes = 76; Total Relationships = 81; Missing Edge Sources = 0.

Layer Contributions and Biomarker Harmonization

Each layer of the graph schema incorporates distinct functional evidence from the course modules:

- **Module 2 Multi-Omics Layer:** Integrates differential expression (RNA and protein log2FC) and GWAS signal strength ($-\log_{10} p$) to establish baseline genetic and transcriptomic evidence for Parkinson's disease (PD).
- **Module 3 Druggability Layer:** Provides machine learning predictions (p_{sm} for small molecules, p_{ab} for antibodies) to quantify targetability.
- **Contextual Functional Layers:** Incorporate phenotypes (HPO), biological pathways (Reactome), lead variants, and clinical candidates (ChEMBL) via Open Targets GraphQL interfaces.
- **Patient Layer:** Adds a synthetic cohort to illustrate individual risk profiles linked via polygenic risk scores (PRS) and risk allele counts.

Attributing evidence directly to edges rather than gene nodes preserves historical and context-dependent provenance. A gene's biological function is static, but its association with a disease or drug varies depending on the study design, tissue type, or clinical trial phase. Placing evidence metrics (e.g., GWAS $-\log_{10} p$, ChEMBL mechanism confidence) on edges prevents information loss when a single gene node interacts with multiple diseases or drugs.

The PAM/CHL1 ambiguity shows that gene symbols are unstable identifiers across public databases. During mapping via Open Targets, PAM resolved ambiguously to MYCBP2, while CHL1 mapped to DDX11 (which lists CHL1 as a historical alias). Enforcing harmonization using official HGNC-approved symbols and cross-referencing against 16 unique Ensembl IDs eliminated dangling edges and ensured 1:1 node-entity mapping across the multi-omics, GWAS, and Open Targets datasets.

Progressive Supranuclear Palsy (PSP) and Alzheimer's disease (AD) were selected as related disease nodes because both exhibit overlapping clinical movement and cognitive features with PD—specifically parkinsonism, bradykinesia, rigidity, postural instability, and depression. Biologically, PSP represents a primary tauopathy sharing pathogenic mechanisms with PD (such as cytoskeletal disruption via MAPT), while AD shares amyloidogenic aggregation pathways (APP/SNCA) and neurodegenerative cascades.

2. Top Targets: Network Centrality and Fused Ranking

Centrality vs. Biological Relevance

The PageRank algorithm evaluates node importance based on the number and quality of incoming directed edges. In this directed knowledge graph, PageRank rewards nodes that act as convergent hubs for variants, synthetic patients, drugs, and pathway annotations.

Gene	PageRank	−log10 p	Druggability	OT PD score	Fused score
FYN	0.017	14.4	0.96	0.452	0.822
SNCA	0.054	169.4	0.023	0.871	0.778
STK39	0.012	41.4	0.925	0.543	0.744
MAPT	0.028	68	0.007	0.678	0.667

The high ranking of SNCA (PageRank = 0.0543) and MAPT (PageRank = 0.0278) aligns with established PD biology. α-synuclein (SNCA) forms insoluble Lewy body aggregates that characterize PD pathology, while tau (MAPT) drives neurofibrillary tangle formation and microtubule instability across parkinsonian syndromes.

However, PageRank alone presents limitations:

1. **Incoming Path Bias:** PageRank heavily rewards node in-degree. SNCA gains substantial network centrality because 7 synthetic patient nodes point to its lead variant (rs356182), which in turn points directly to SNCA ((Patient)-[:CARRIES]->(Variant)-[:OCCURS_IN]->(Gene)). This patient cohort connection artificially inflates SNCA's raw score relative to targets that lack patient-level connectivity.
2. **Structural Sinks and Node Ties:** Eleven pipeline genes tie at a minimum PageRank score of 0.0117. In a directed graph, nodes that primarily point *outward* to disease or pathway nodes rather than receiving incoming edges from multiple upstream entities act as network sinks, receiving only the baseline damping factor distribution $(1 - d)$.

A key non-obvious hit highlighted by the fused score (mean percentile rank across PageRank, GWAS strength, and Module 3 druggability) is **FYN** (Fused Score = 0.822). Although SNCA holds the highest PageRank and GWAS significance ($-\log_{10} p = 169.4$), its druggability is poor (0.023). Conversely, FYN achieves a high fused score due to its high druggability prediction (0.960 for small-molecule inhibition) combined with strong network topology and GWAS significance ($-\log_{10} p = 14.4$). Similarly, **STK39** (Fused Score = 0.744) ranks above MAPT (0.667) due to high druggability (0.925) and strong GWAS evidence ($-\log_{10} p = 41.4$), despite lacking Reactome pathway annotations.

The Druggability vs. Clinical Drug Tension

A clear tension exists for MAPT: it has three anti-tau monoclonal antibodies in Phase II clinical trials (Gosuranemab, Tilavonemab, Bepranemab), yet its Module 3 druggability classifier predicts a score near zero (0.0069).

This discrepancy stems from the training baseline of the Module 3 machine learning classifier. Druggability models trained on conventional structural, sequence, and biochemical features reward targets with well-defined hydrophobic binding pockets typical of small-molecule enzymes, kinases, or G-protein coupled receptors (GPCRs). Because MAPT encodes an intrinsically disordered protein (tau) lacking a stable 3D catalytic pocket, the classifier assigns it a near-zero small-molecule druggability score. The existence of antibodies in clinical trials highlights a modality mismatch: immunotherapies target extracellular or non-traditional structural surfaces via protein-protein binding, a mechanism that standard pocket-based druggability classifiers are not trained to identify.

Community Interpretation

Modularity clustering using greedy community detection grouped densely connected nodes into functional modules.

- **Community 1 Analysis (Proteinopathy Module):**
  - *Members:* SNCA, MAPT, Prasinezumab, Cinpanemab, Gosuranemab, Tilavonemab, Bepranemab, *Amyloid fiber formation*, *PKR-mediated signaling*, *Activation of AMPK downstream of NMDARs*, rs17649553.
  - *Interpretation:* Community 1 functions as a true **proteinopathy and neurodegeneration module** rather than a simple computational artifact of "genes with drugs." While therapeutic antibodies link SNCA and MAPT, their shared involvement in cytotoxic protein aggregation (*Amyloid fiber formation*) and stress-response cascades (*PKR-mediated signaling*) establishes a biological subnetwork centered on pathological protein misfolding.

3. Hypothesis Generation from Repurposing Paths

Candidate Pathway Traversal

Traversing the 5-hop cross-disease path reveals a mechanism for therapeutic repurposing:

$$\text{(Gosuranemab)} \xrightarrow{\text{TARGETS}} \text{(MAPT)} \xrightarrow{\text{ASSOCIATED_WITH}} \text{(PSP)} \xrightarrow{\text{HAS_PHENOTYPE}} \text{(Parkinsonism)} \xleftarrow{\text{HAS_PHENOTYPE}} \text{(Parkinson's disease)}$$

Formulated Hypothesis

Targeting extracellular tau clearance via anti-tau monoclonal antibody therapy (e.g., Gosuranemab) in patients presenting with parkinsonian motor phenotypes is consistent with shared tau-mediated pathophysiological cascades between Progressive Supranuclear Palsy and Parkinson's disease, suggesting that anti-tau biologics may attenuate secondary tauopathy in PD sub-phenotypes.

Experimental Validation Plan

To test this hypothesis *in vitro*:

1. **Model:** Human induced pluripotent stem cell (iPSC)-derived dopaminergic neurons harboring the SNCA A53T mutation or exposed to pre-formed α-synuclein fibrils (PFFs) to induce cross-seeded tau aggregation.
2. **Assay & Treatment:** Apply Gosuranemab or an isotype control antibody to the culture media across a dose range (0.1–10 μg/mL) over a 14-day incubation period.
3. **Readout:** Quantify intracellular hyperphosphorylated tau (p-tau181 / p-tau231) and insoluble tau oligomers using TR-FRET and Western blot. Assess neuronal survival and synaptic density using high-content imaging for MAP2 and PSD-95 markers.

Clinical Context & Utility

Clinically, Phase II trials of anti-tau antibodies (e.g., Gosuranemab, Tilavonemab) failed to achieve primary efficacy endpoints in PSP. However, the path remains biologically informative for two reasons:

1. **Efficacy vs. Target Engagement:** The clinical failure of Gosuranemab in PSP was primarily driven by inadequate microglial uptake and insufficient target engagement against intracellular non-N-terminal tau species, rather than an invalidation of the underlying tau hypothesis.
2. **Subtype Stratification:** The network path highlights phenotype-driven shared pathways. In PD, where tau pathology acts as a modifier rather than the primary driver, anti-tau therapeutics may show utility when deployed as combination therapies in stratified patient subsets exhibiting elevated CSF p-tau markers or high polygenic risk for tauopathy.

4. Critical Evaluation of Knowledge Graph Limitations

Knowledge graphs structure biological hypotheses, but their predictions are subject to data noise, missing nodes, and structural assumptions.

Weak point	Example	Effect on results
Sub-genome-wide gene-PD edges	CHL1 (−log10 p 5.4), WDR1 (6.1), PPP6R2 (7.2)	May add false-positive gene nodes
No Open Targets PD association	PBXIP1 (−log10 p 38.7), WDR1	GWAS strength alone does not identify the effector gene
No Reactome annotation	PAM, PBXIP1, PPP6R2, STK39	No mechanistic context; cluster with the disease hub (Community
Polypharmacology	Dasatinib → FYN (1 of 10 SRC-family targets)	Any effect may be off-target
Positional variant mapping	rs356182 → SNCA	Mapped gene may not be the causal gene
Synthetic patients	12 patients, 7 CARRIES edges	Inflate SNCA PageRank; demonstrate the query only
Missing genes	GBA1, LRRK2	Dropped in Module 2 filtering; key PD genes absent
Incomplete drug layer	NDUFAF2 (metformin, ME-344 omitted)	Complex-I-wide inhibitors excluded

Omitting Primary PD Drivers (GBA1, LRRK2): GBA1 and LRRK2—the two most prominent monogenic risk genes for PD—were omitted due to the strict inner join applied across the Module 2 multi-omics pipeline (requiring concurrent RNA, protein, and GWAS data). This highlights how stringent data filtering can introduce selection bias, removing top-tier drug targets from downstream graph analysis.

Impact of Filtering Standards

When re-querying the 2-hop drug target network ((Drug)-[:TARGETS]->(Gene)-[:ASSOCIATED_WITH]->(Disease)) using a strict genome-wide evidence filter (WHERE a.confidence = 'genome-wide'), the query returned the exact same 6 drug candidates:

The strict threshold produced an identical drug candidate list because the initial ChEMBL/Open Targets drug mapping only retrieved compounds linked to top-tier target genes (SNCA, MAPT, FYN), all of which independently exceeded the genome-wide significance threshold ($-\log_{10} p \geq 7.30$). The sub-genome-wide targets (CHL1, WDR1, PPP6R2) lacked Phase II/III drug candidates in ChEMBL. While the strict threshold confirmed the robust genetic backing of the primary drug targets, it did not filter out polypharmacological ambiguities—such as Dasatinib, which targets FYN as one of 10 SRC-family kinases.

Database State Management

During development, a stale Neo4j instance containing leftover nodes from a previous Alzheimer's disease graph altered query outputs, producing duplicate node counts and false-positive path connections. To ensure reproducible results, the execution script was modified to clear the graph database before loading data:

Cypher

```
MATCH (n) DETACH DELETE n;
```

Running this explicit reset prior to UNWIND ... MERGE ingestion ensured clean node/edge execution and exact count matches (§1: 76 nodes, 81 edges).

5. Methodology & AI Transparency Report

AI use.

An AI assistant queried the Open Targets Platform to retrieve Ensembl IDs, HPO phenotypes, Reactome pathways, ChEMBL drug mechanisms, and association scores, and drafted the §1 data cell and the added check and query cells. It also proposed several curation choices: the pathway inclusion rule, PSP and Alzheimer's as related diseases, the drug list, the synthetic cohort, and the fused-score formula. I verified the build with the harmonization cell (16 genes → 16 unique Ensembl IDs, 0 dangling edges), the source check (0 edges missing a source), and matching §1 and §2 counts (76 nodes, 81 relationships), and I read the query outputs against my input files.

Verification and Integrity Checks

To ensure data integrity, the following manual and programmatic checks were executed:

1. **Harmonization Check Cell:** Executed programmatic assertions verifying that all 16 gene symbols mapped to 16 unique Ensembl IDs, with zero dangling edge endpoints.
2. **Source Completeness Audit:** Ran Python validation scripts confirming that every edge dictionary contained a populated source key, returning 0 missing edge sources.
3. **Cross-Section Reconciliation:** Verified that node ($n = 76$) and edge ($n = 81$) counts in §1 matched the Neo4j instance load counts in §2 (MATCH (n) RETURN count(*)).
4. **Source-Data Audit:** Re-read Cypher query outputs directly against raw inputs in targets_parkinsons.csv, targets_parkinsons_w3.csv, and pd_gwas_associations.tsv to verify target scores, $-\log_{10} p$ calculations, and drug mechanisms.

6. Knowledge Graph Visualization:

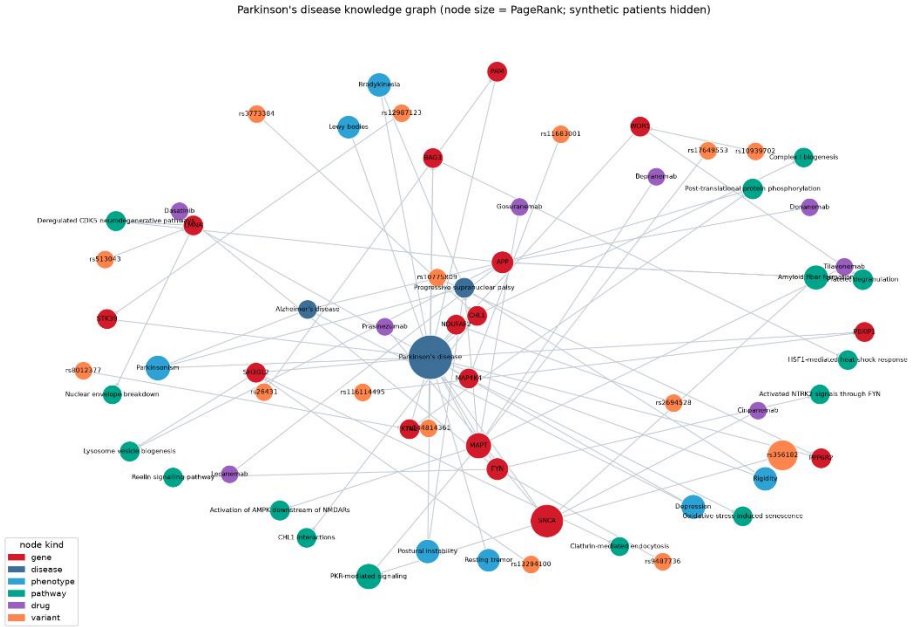


Figure 1: *Parkinson's disease knowledge graph (76 nodes, 81 relationships; synthetic patient nodes hidden for visual clarity). Node colour indicates entity type (gene = red, disease = dark blue, phenotype = light blue, pathway = teal, drug = purple, variant = orange)*